

WASP-11b

R-1079

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-10_151205 · created 2026-07-30T14:57:33+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2457357.9794 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.162 +/- 0.052
Transit depth (Rp/Rs)^2	2.63 +/- 1.69 [%]
Orbital Inclination (inc)	85.65 +/- 1.68
Airmass coefficient 1 (a1)	2.01 +/- 1.15
Airmass coefficient 2 (a2)	-0.67 +/- 1.48
Scatter in the residuals of the lightcurve fit is	52.86 %
Best Comparison Star	None
Optimal Aperture	2.53
Optimal Annulus	7.0
Transit Duration (day)	0.019 +/- 0.015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.162 ± 0.052 vs 0.131 (deviation 0.6σ)
Duration fit vs published	0.019 d vs 0.1075 d (deviation 5.9σ)
Mid-time O–C	-5358.1 min
Depth SNR	3.1
Residual scatter	52.86 %

Light curve

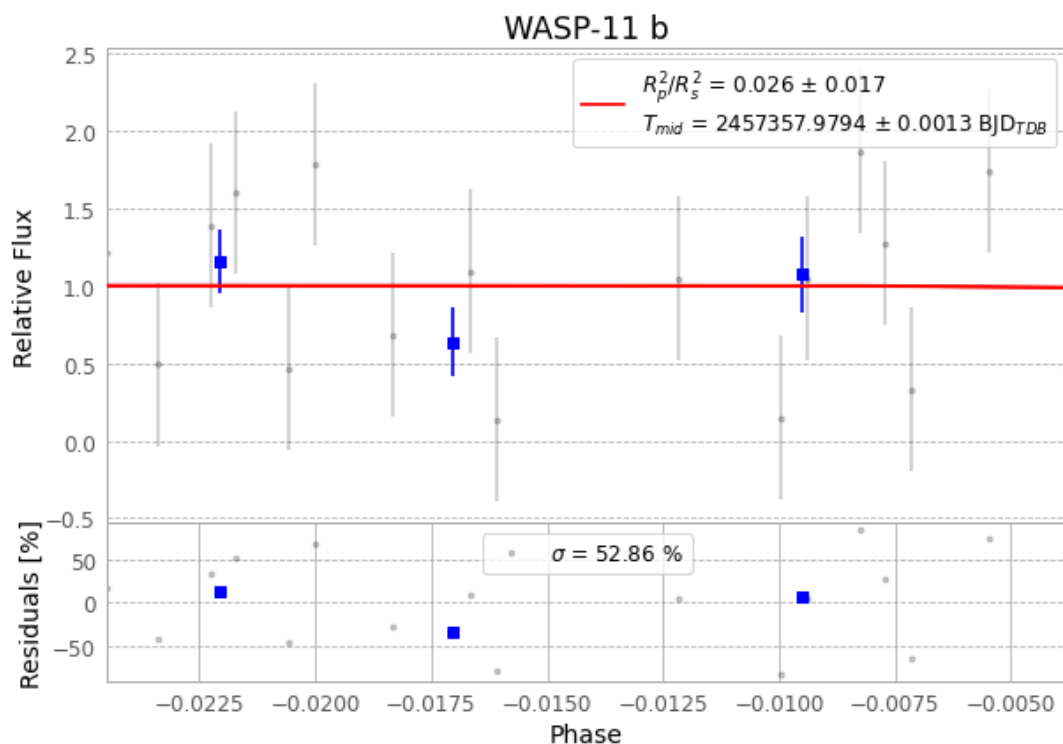


Figure 1 — FinalLightCurve_WASP-11 b_2015-12-04.png (SHA-256 b5381e36943d0654...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [340, 476], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 effe61c11068794c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

94 FITS frames (62 MB), HATP-10151205021812.FITS → HATP-10151205065712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-151205.log (05976245da09...), FinalLightCurve_WASP-11 b_2015-12-04.png (b5381e36943d...), FinalLightCurve_WASP-11 b_2015-12-04.csv (b526946fc904...)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 14:57Z.